Pilot experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Pilot experiments before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Predefining an analysis plan

Without opening the old lesson, explain one key idea from Predefining an analysis plan. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Research ethics and consent

Without opening the old lesson, explain one key idea from Research ethics and consent. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Ablation studies

Without opening the old lesson, explain one key idea from Ablation studies. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Pilot experiments trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Pilot** experiments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Pilot experiments explanation against the supplied facts before discussing it.

Use the worked case for **Pilot** experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

> **Pilot** six cases exposes ambiguous labels in two. What should happen before the primary study?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Pilot experiments facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Pilot experiments facts and keep your reasoning.

Pilot five cases reveals wrong units in duration logs; team corrects units and reports the **pilot** as final **evidence**. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Pilot experiments facts and keep your reasoning.

A report omits this qualification: A **pilot** may assess **feasibility** but is not automatically powered or independent **evidence** for the final hypothesis.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Pilot experiments facts and keep your reasoning.

> **Pilot** six cases exposes ambiguous labels in two. What should happen before the primary study?

My reasoning and evidence

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Pilot experiments facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Pilot experiments to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled **pilot** experiments **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **pilot** experiments.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Pilot experiments

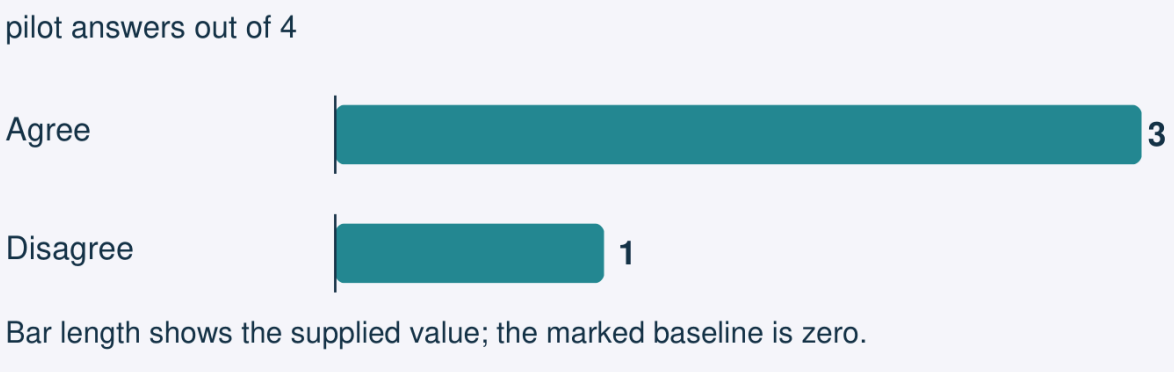
> A pilot tests the procedure. A pilot checks whether the team can score answers consistently before running the main study. Explain the labelled step before checking the changed condition.

A **pilot** tests the procedure. A **pilot** checks whether the team can score answers consistently before running the main study.

Two reviewers score four practice answers. They agree on three and disagree on one; these answers are excluded from final evaluation.

AI ACADEMY / AI-7/22

Reviewer agreement on four pilot answers



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/22

Pilot experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Try the workflow

Both reviewers apply the draft rubric to four practice answers.



2. Measure agreement

Agreement is 3/4=75% on this pilot.



3. Repair the rubric

Discuss the disagreement and clarify the scoring rule before the main run.

CHANGE ONE THING

Add the corrected pilot answers into final evaluation.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Calculate reviewer agreement on the four **pilot** answers. What can the disagreement improve, and why are those revised practice answers not untouched final **evidence**?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A **pilot** tests the procedure. A **pilot** checks whether the team can score answers consistently before running the main study.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Discuss the disagreement and clarify the scoring rule before the main run.

1. Try the workflow

2. Repair the rubric

3. Measure agreement

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Repair the rubric Discuss the disagreement and clarify the scoring rule before the main run.

A **pilot** can reveal missing instructions, practical delays or ambiguous scoring. It is preparation, not an untouched final estimate after the team uses it to revise methods.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Add the corrected **pilot** answers into final evaluation. Which conclusion is supported by the supplied facts?

1. Declare them previously exposed rather than untouched final **evidence**.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Declare them previously exposed rather than untouched final **evidence**. Rubric development already used those answers.

Repair a missing building block

Review Predefining an analysis plan (ai-7/21). Goal: Explain the **mechanism** and terminology of predefining an analysis plan. Short practice: Apply the recorded exclusion rule.

Can two readable model failures be removed, and how must a necessary later plan change be disclosed?. Return to this week and explain what you changed.

Review Research ethics and consent (ai-7/19). Goal: Explain the **mechanism** and terminology of research ethics and consent. Short practice: Identify which uses the drawing consent permits and excludes.

Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?. Return to this week and explain what you changed.

Review Ablation studies (ai-7/16). Goal: Explain the **mechanism** and terminology of ablation studies. Short practice: Calculate the filter ablation's two rates and count change.

Which single component is altered, and why does also changing the model prevent isolating the filter?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

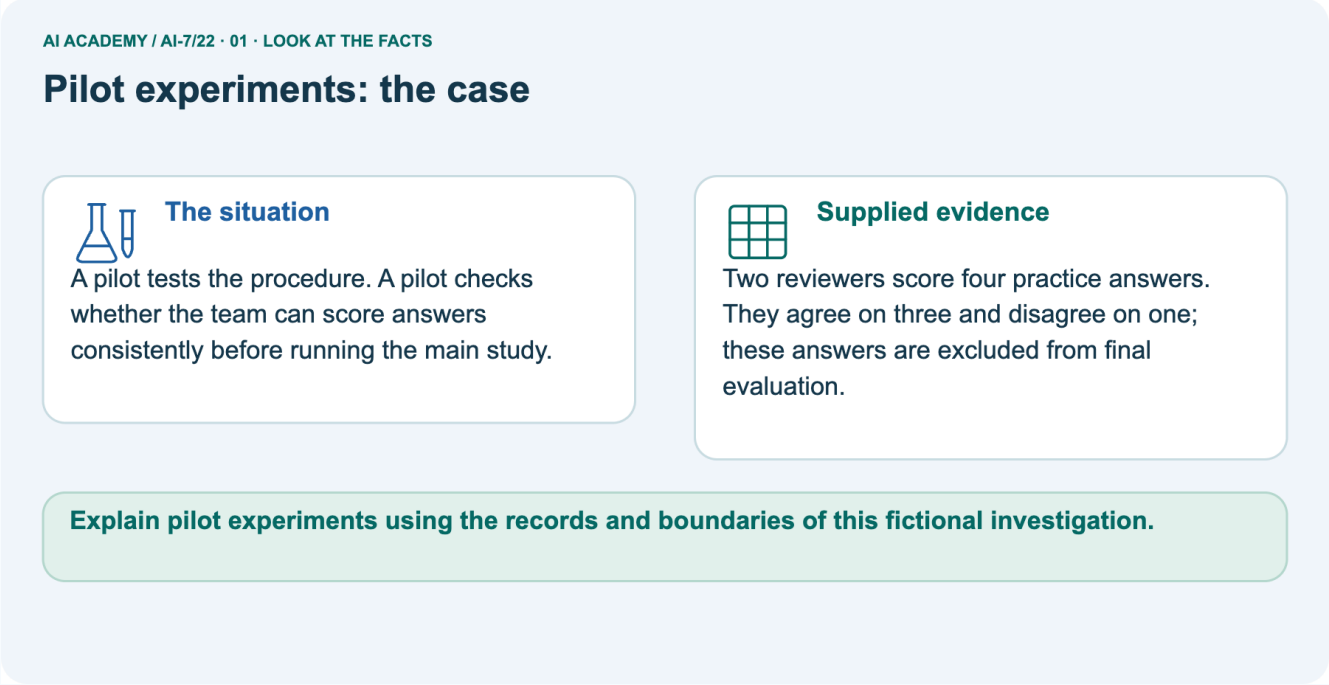
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Pilot experiments picture from its labels, then record a relevant limit.

01. Pilot experiments: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

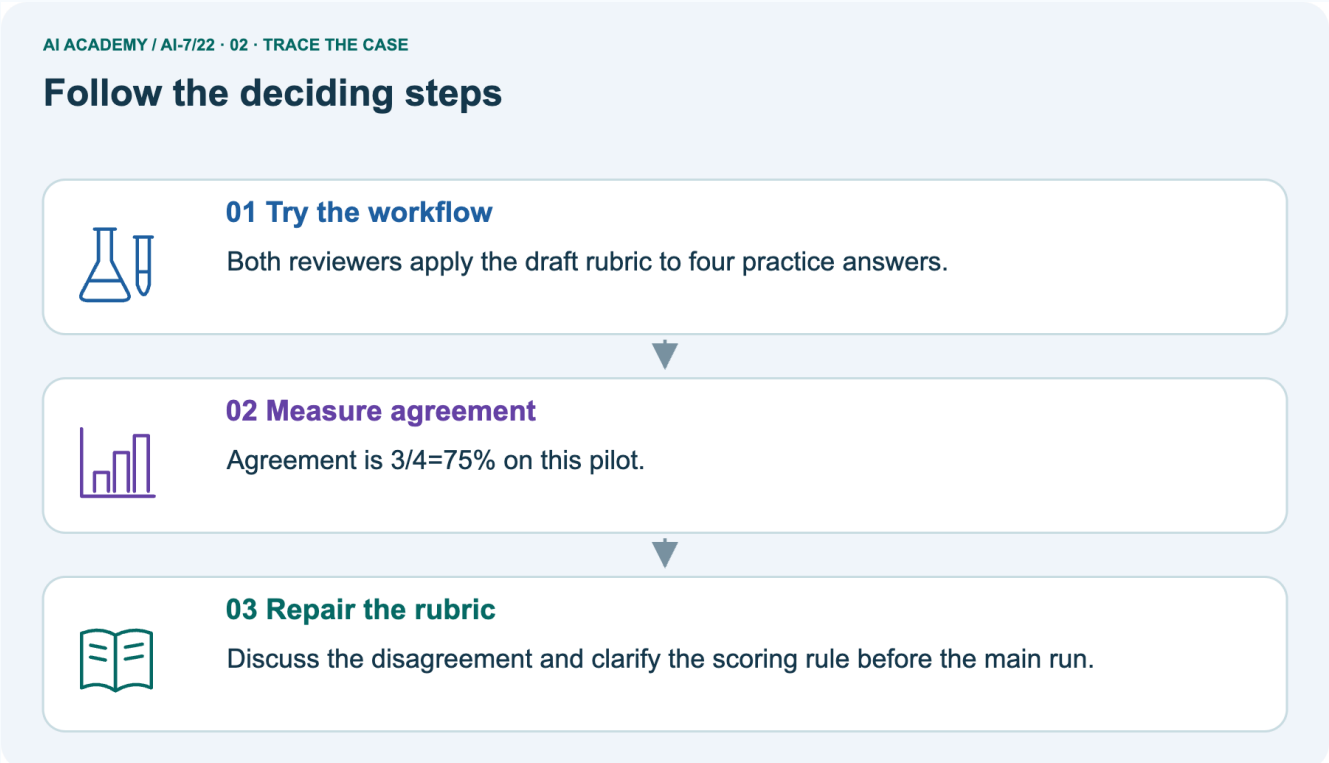
A **pilot** tests the procedure. A **pilot** checks whether the team can score answers consistently before running the main study. Supplied facts: Two reviewers score four practice answers.

They agree on three and disagree on one; these answers are excluded from final evaluation.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



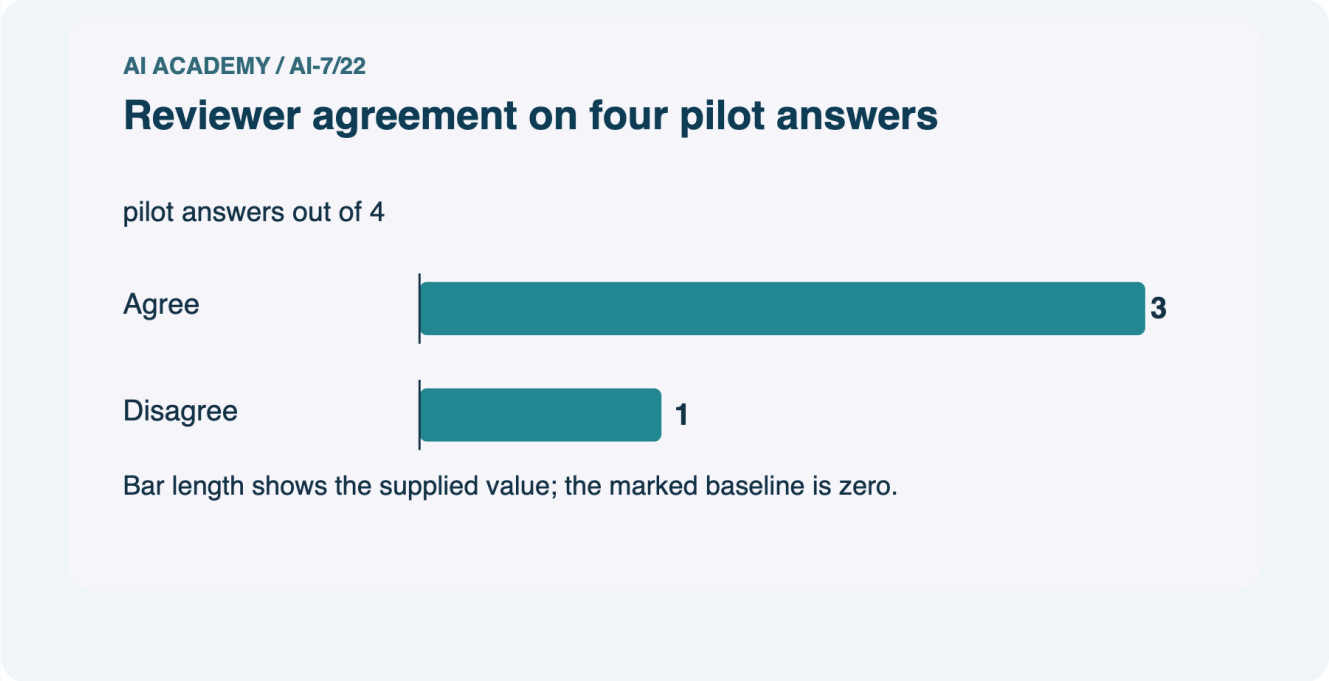
Follow the labelled picture with the matching explanation. Open the vector diagram

Try the workflow: Both reviewers apply the draft rubric to four practice answers. → Measure agreement: Agreement is 3/4=75% on this **pilot**. → Repair the rubric: Discuss the disagreement and clarify the scoring rule before the main run.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Reviewer agreement on four pilot answers



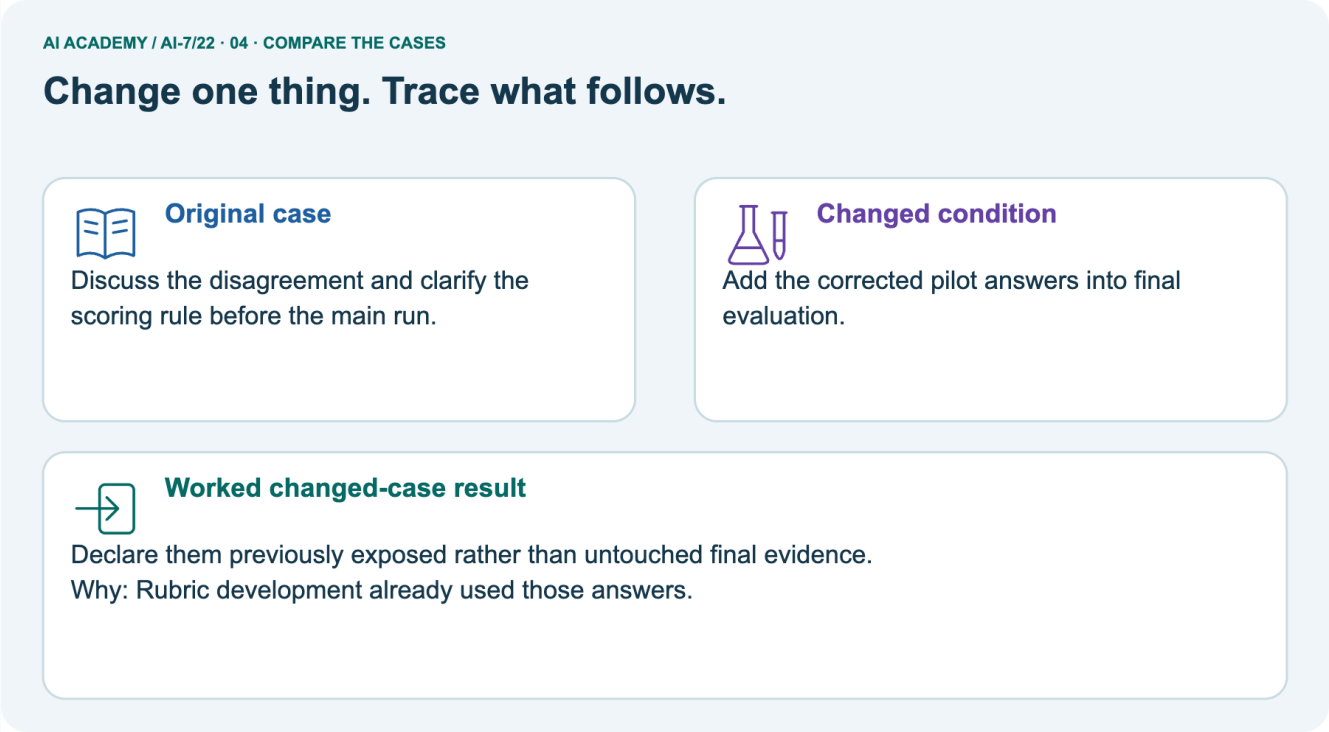
Follow the labelled picture with the matching explanation. Open the vector diagram

Reviewer agreement on four **pilot** answers. Units: **pilot** answers out of 4. Agree: 3. Disagree: 1.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

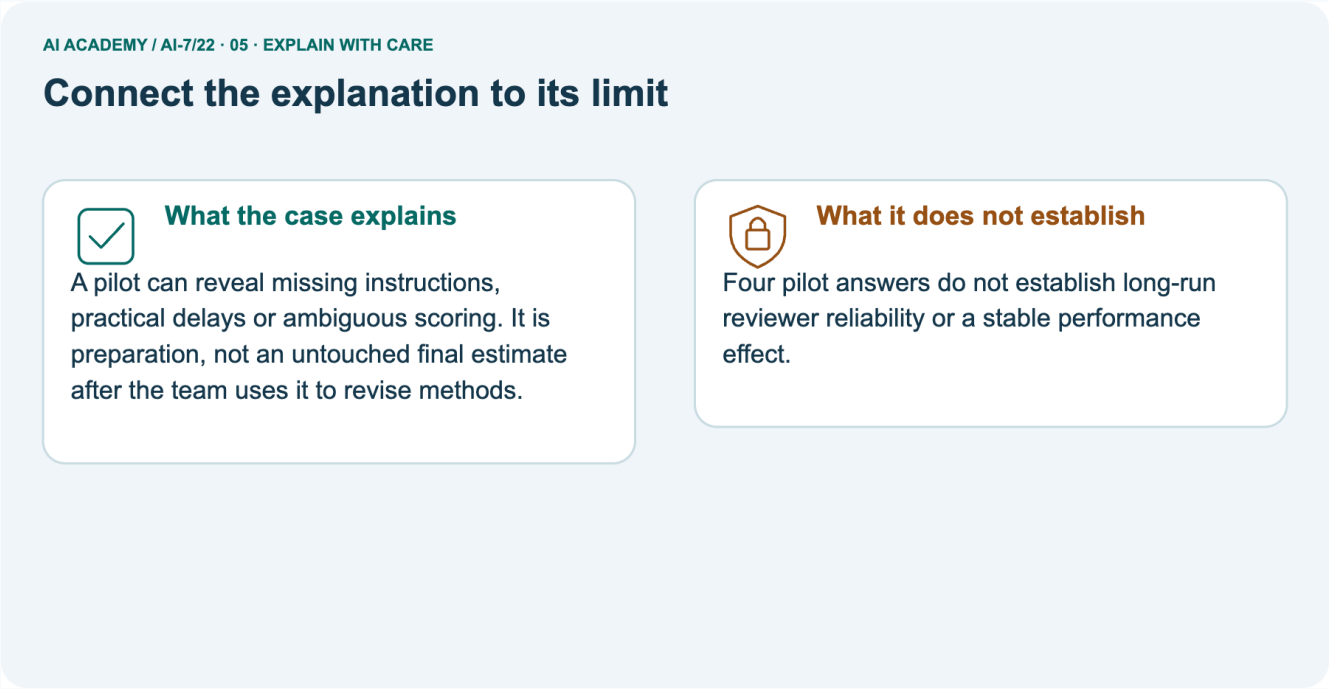
Original case: Discuss the disagreement and clarify the scoring rule before the main run. Changed condition: Add the corrected **pilot** answers into final evaluation.

Expected result: Declare them previously exposed rather than untouched final **evidence**. Reason: Rubric development already used those answers.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

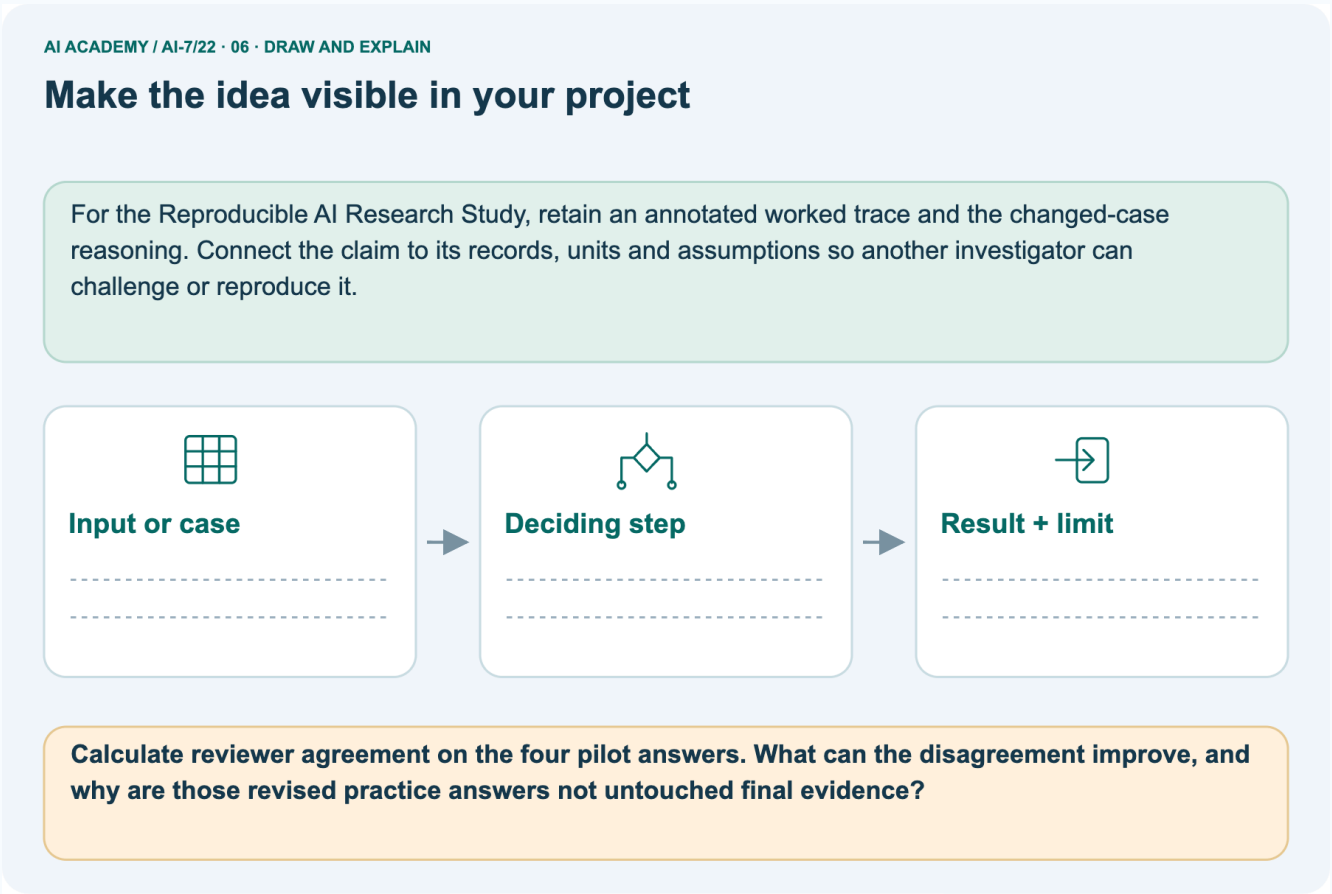
A **pilot** can reveal missing instructions, practical delays or ambiguous scoring. It is preparation, not an untouched final estimate after the team uses it to revise methods.

Limit: Four **pilot** answers do not establish long-run reviewer reliability or a stable performance effect.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate reviewer agreement on the four **pilot** answers.

What can the disagreement improve, and why are those revised practice answers not untouched final **evidence**?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.